ASTANA IT UNIVERSITY

Assignment 4

ADA-2601M-2603M

Part 1. The Laplace transform

1. Find $L[\sin(2t+3)]$
2. Find $L[sin4t - e^{-5t}]$
3. Find $L[\frac{sin^2 t}{t}]$
4. Prove that: **If $\boldsymbol{L[f(t)] = F(s)}$, then $\boldsymbol{L[e^{-at} \cdot f(t)] = F(s+a)}$.**
5. Find $L[t \cdot e^{2t}]$
6. Find $L[t^5 \cdot e^{-t}]$
7. Find $L[e^{-2t} \cdot sin3t]$

Part 2. Fourier series

Expand a function in a Fourier series on the interval (−π,π):

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 \le x < \pi \end{cases}$$